

Database Technologies – Assignment

To solve below queries use “hr” database

- Write a query to get unique department ID from employee table.

SELECT DISTINCT DEPARTMENT_ID FROM EMPLOYEES;

```
W2_99546_Pratiksha>SELECT DISTINCT DEPARTMENT_ID FROM EMPLOYEES;
```

DEPARTMENT_ID
NULL
10
20
30
40
50
60
70
80
90
100
110

- Write a query to get all the employee's details from the employees table displaying the first names in descending order.

Select *FROM employees ORDER BY FIRST_NAME DESC ;

```
W2_99546_Pratiksha>Select *FROM employees ORDER BY FIRST_NAME DESC;
```

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
180	Winston	Taylor	WTAYLOR	1.650.555.0145	2016-01-24	SH_CLERK	3200.00	NULL	120	50
171	William	Smith	WSMITH	44.1632.960026	2017-02-23	SA_REP	7400.00	0.15	148	80
206	William	Gietz	WGIEZT	1.515.555.0171	2012-06-07	AC_ACCOUNT	8300.00	NULL	205	110
195	Vance	Jones	VJONES	1.650.555.0160	2017-03-17	SH_CLERK	2800.00	NULL	123	50
106	Valli	Jackson	VJACKSON	1.590.555.0106	2016-02-05	IT_PROG	4800.00	NULL	103	60
141	Trenna	Rajs	TRAJS	1.650.555.0141	2013-10-17	ST_CLERK	3500.00	NULL	124	50
132	TJ	Olson	TJOLSON	1.650.555.0132	2017-04-10	ST_CLERK	2100.00	NULL	121	50
190	Timothy	Venzl	TVENZL	1.650.555.0155	2016-07-11	SH_CLERK	2900.00	NULL	122	50
170	Taylor	Fox	TFOX	44.1632.960025	2016-01-24	SA_REP	9600.00	0.20	148	80
203	Susan	Jacobs	SJACOBS	1.515.555.0168	2012-06-07	HR_REP	6500.00	NULL	101	40
173	Sundita	Kumar	SKUMAR	44.1632.960028	2018-04-21	SA_REP	6100.00	0.10	148	80
166	Sundar	Ande	SANDE	44.1632.960021	2018-03-24	SA_REP	6400.00	0.10	147	80
100	Steven	King	SKING	1.515.555.0100	2013-06-17	AD_PRES	24000.00	NULL	NULL	90
127	Anthony	Deveau	ADEVEAU	1.500.555.0122	2017-02-07	SH_CLERK	3900.00	NULL	121	50
167	Amit	Banda	ABANDA	44.1632.960022	2018-04-21	SA_REP	6200.00	0.10	147	80
175	Alyssa	Hutton	AHUTTON	44.1632.960030	2018-03-19	SA_REP	8800.00	0.25	149	80
158	Allan	McEwen	AMCEWEN	44.1632.960013	2014-08-01	SA_REP	9000.00	0.35	146	80
185	Alexis	Bull	ABULL	1.650.555.0150	2015-02-20	SH_CLERK	4100.00	NULL	121	50
103	Alexander	James	AJAMES	1.590.555.0103	2016-01-03	IT_PROG	9000.00	NULL	102	60
115	Alexander	Khoo	AKHOO	1.515.555.0115	2013-05-18	PU_CLERK	3100.00	NULL	114	30
147	Alberto	Errazuriz	AERRAZUR	44.1632.960002	2015-03-10	SA_MAN	12000.00	0.30	100	80
196	Alana	Walsh	AWALSH	1.650.555.0161	2016-04-24	SH_CLERK	3100.00	NULL	124	50
121	Adam	Fripp	AFRIPP	1.650.555.0121	2015-04-10	ST_MAN	8200.00	NULL	100	50

- Write a query to get the employee ID, names (first_name and last_name), salary in ascending order of salary.

SELECT EMPLOYEE_ID, FIRST_NAME, LAST_NAME, SALARY
FROM employees
ORDER BY SALARY ASC;

```
W2_99546_Praticrsha>SELECT EMPLOYEE_ID, FIRST_NAME, LAST_NAME, SALARY
-> FROM employees
-> ORDER BY SALARY ASC;
```

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	SALARY
132	TJ	Olson	2100.00
128	Steven	Markle	2200.00
136	Hazel	Philtanker	2200.00
127	James	Landry	2400.00
135	Ki	Gee	2400.00
119	Karen	Colmenares	2500.00
131	James	Marlow	2500.00
140	Joshua	Patel	2500.00
144	Peter	Vargas	2500.00
182	Martha	Sullivan	2500.00
191	Randall	Perkins	2500.00
118	Guy	Himuro	2600.00
143	Randall	Matos	2600.00
198	Donald	OConnell	2600.00
199	Douglas	Grant	2600.00
126	Irene	Mikkilineni	2700.00
139	John	Seo	2700.00
149	Eleni	Zlotkey	10500.00
162	Clara	Vishney	10500.00
114	Den	Li	11000.00
148	Gerald	Cambrault	11000.00
174	Ellen	Abel	11000.00
168	Lisa	Ozer	11500.00
147	Alberto	Errazuriz	12000.00
108	Nancy	Gruenberg	12008.00
205	Shelley	Higgins	12008.00
201	Michael	Martinez	13000.00
146	Karen	Partners	13500.00
145	John	Singh	14000.00
101	Neena	Yang	17000.00
102	Lex	Garcia	17000.00
100	Steven	King	24000.00

4. Display first name and joining date of the employees who are either IT Programmer or Sales Man.

```
SELECT FIRST_NAME, HIRE_DATE, JOB_ID
```

```
FROM employees
```

```
WHERE JOB_ID='IT_PROG'
```

```
OR
```

```
JOB_ID='SA_MAN';
```

```

W2_99546_Pratiksha>SELECT FIRST_NAME, HIRE_DATE, JOB_ID
-> FROM employees
-> WHERE JOB_ID='IT_PROG'
-> OR
-> JOB_ID ='SA_MAN';

```

FIRST_NAME	HIRE_DATE	JOB_ID
Alexander	2016-01-03	IT_PROG
Bruce	2017-05-21	IT_PROG
David	2015-06-25	IT_PROG
Valli	2016-02-05	IT_PROG
Diana	2017-02-07	IT_PROG
John	2014-10-01	SA_MAN
Karen	2015-01-05	SA_MAN
Alberto	2015-03-10	SA_MAN
Gerald	2017-10-15	SA_MAN
Eleni	2018-01-29	SA_MAN

5.Display details of employees with employee ID 150 or 160.

```

Select * FROM employees
WHERE EMPLOYEE_ID IN (150 , 160);

```

```
W2_99546_Pratiksha>Select * FROM employees
-> WHERE EMPLOYEE_ID IN (150 , 160);
```

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
150	Sean	Tucker	STUCKER	44.1632.960005	2015-01-30	SA_REP	10000.00	0.30	145	80
160	Louise	Doran	LDORAN	44.1632.960015	2015-12-15	SA_REP	7500.00	0.30	146	80

6.Display first name, salary, commission pct, and hire date for employees with salary less than 10000.

```

SELECT FIRST_NAME, COMMISSION_PCT, HIRE_DATE, SALARY
FROM employees
WHERE SALARY<10000;

```

```
W2_99546_Pratiksha>SELECT FIRST_NAME, COMMISSION_PCT, HIRE_DATE,SALARY
-> FROM employees
-> WHERE SALARY<10000;
```

FIRST_NAME	COMMISSION_PCT	HIRE_DATE	SALARY
Alexander	NULL	2016-01-03	9000.00
Bruce	NULL	2017-05-21	6000.00
David	NULL	2015-06-25	4800.00
Valli	NULL	2016-02-05	4800.00
Diana	NULL	2017-02-07	4200.00
Daniel	NULL	2012-08-16	9000.00
John	NULL	2015-09-28	8200.00
Ismael	NULL	2015-09-30	7700.00
Jose Manuel	NULL	2016-03-07	7800.00
Luis	NULL	2017-12-07	6900.00
Alexander	NULL	2013-05-18	3100.00
Shelli	NULL	2015-12-24	2900.00
Sigal	NULL	2015-07-24	2800.00
Guy	NULL	2016-11-15	2600.00
Karen	NULL	2017-08-10	2500.00
Matthew	NULL	2014-07-18	8000.00
Adam	NULL	2015-04-10	8200.00
Payam	NULL	2013-05-01	7900.00
Shanta	NULL	2015-10-10	6500.00
Kevin	NULL	2017-11-16	5800.00
Julia	NULL	2015-07-16	3200.00
Tanner	NULL	2016-09-28	2700.00
Vance	NULL	2017-03-17	2800.00
Alana	NULL	2016-04-24	3100.00
Kevin	NULL	2016-05-23	3000.00
Donald	NULL	2017-06-21	2600.00
Douglas	NULL	2018-01-13	2600.00
Jennifer	NULL	2013-09-17	4400.00
Pat	NULL	2015-08-17	6000.00
Susan	NULL	2012-06-07	6500.00
William	NULL	2012-06-07	8300.00

7. Display employees where the first name or last name starts with S.

```
SELECT FIRST_NAME, LAST_NAME
```

```
FROM employees
```

```
WHERE FIRST_NAME LIKE 's%'
```

```
OR
```

```
LAST_NAME LIKE 's%';
```

```

W2_99546_Pratiksha>SELECT FIRST_NAME, LAST_NAME
-> FROM employees
-> WHERE FIRST_NAME LIKE 's%'
-> OR
-> LAST_NAME LIKE 's%';
+-----+-----+
| FIRST_NAME | LAST_NAME |
+-----+-----+
| Sundar     | Ande      |
| Shelli     | Baida     |
| Sarah      | Bell      |
| Shelley    | Higgins   |
| Susan      | Jacobs    |
| Steven     | King      |
| Sundita    | Kumar     |
| Steven     | Markle    |
| Samuel     | McLeod    |
| Nandita    | Sarchand  |
| Ismael     | Sciarra   |
| John       | Seo       |
| Sarath     | Sewall    |
| John       | Singh     |
| Lindsey    | Smith     |
| William    | Smith     |
| Stephen    | Stiles    |
| Martha     | Sullivan  |
| Patrick    | Sully     |
| Sigal      | Tobias    |
| Sean       | Tucker    |
| Shanta     | Vollman   |
+-----+-----+
22 rows in set (0.00 sec)

```

8. Display details of jobs in the descending order of the title.

```

SELECT job_title, job_ID, MIN_SALARY, MAX_SALARY
FROM jobs
ORDER BY job_title DESC;

```

```

W2_99546_Pratiksha>SELECT job_title, job_ID, MIN_SALARY, MAX_SALARY
-> FROM jobs
-> ORDER BY job_title DESC;

```

job_title	job_ID	MIN_SALARY	MAX_SALARY
Stock Manager	ST_MAN	5500	8500
Stock Clerk	ST_CLERK	2008	5000
Shipping Clerk	SH_CLERK	2500	5500
Sales Representative	SA_REP	6000	12008
Sales Manager	SA_MAN	10000	20080
Purchasing Manager	PU_MAN	8000	15000
Purchasing Clerk	PU_CLERK	2500	5500
Public Relations Representative	PR_REP	4500	10500
Public Accountant	AC_ACCOUNT	4200	9000
Programmer	IT_PROG	4000	10000
President	AD_PRES	20080	40000
Marketing Representative	MK_REP	4000	9000
Marketing Manager	MK_MAN	9000	15000
Human Resources Representative	HR_REP	4000	9000
Finance Manager	FI_MGR	8200	16000
Administration Vice President	AD_VP	15000	30000
Administration Assistant	AD_ASST	3000	6000
Accounting Manager	AC_MGR	8200	16000
Accountant	FI_ACCOUNT	4200	9000

```

19 rows in set (0.00 sec)

```

9. Display the details of the employees from department 30, earning the salary greater 10000 but do not receive the commission.

```

SELECT EMPLOYEE_ID, SALARY, FIRST_NAME, LAST_NAME, EMAIL, PHONE_NUMBER, HIRE_DATE,
JOB_ID, COMMISSION_PCT, MANAGER_ID, DEPARTMENT_ID

```

```

FROM employees

```

```

WHERE DEPARTMENT_ID=30

```

```

AND

```

```

SALARY>10000

```

```

AND

```

```

COMMISSION_PCT IS NULL;

```

```

W2_99546_Pratiksha>SELECT EMPLOYEE_ID, SALARY, FIRST_NAME, LAST_NAME, EMAIL, PHONE_NUMBER, HIRE_DATE, JOB_ID, COMMISSION_PCT, MANAGER_ID, DEPARTMENT_ID
-> FROM employees
-> WHERE DEPARTMENT_ID=30
-> AND
-> AND
-> SALARY>10000
-> AND
-> COMMISSION_PCT IS NULL;

```

EMPLOYEE_ID	SALARY	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
114	11000.00	Den	Li	DLT	1.515.555.0114	2012-12-07	PU_MAN	NULL	100	30

```

1 row in set (0.02 sec)

```

10. Display the details of employees reporting to the managers with employee id 100 and 120.

```

SELECT EMPLOYEE_ID, SALARY, FIRST_NAME, LAST_NAME, EMAIL, PHONE_NUMBER, HIRE_DATE,
JOB_ID, COMMISSION_PCT, MANAGER_ID, DEPARTMENT_ID

```

```

FROM employees

```

```

WHERE MANAGER_ID IN (100,120);

```

```

W2_99546_Pratiksha>SELECT EMPLOYEE_ID, SALARY, FIRST_NAME, LAST_NAME, EMAIL, PHONE_NUMBER, HIRE_DATE, JOB_ID, COMMISSION_PCT, MANAGER_ID, DEPARTMENT_ID
-> FROM employees
-> WHERE MANAGER_ID IN (100,120);

```

EMPLOYEE_ID	SALARY	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
101	17000.00	Neena	Yang	NYANG	1.515.555.0101	2015-09-21	AD_VP	NULL	100	90
102	17000.00	Lex	Garcia	LGARCIA	1.515.555.0102	2011-01-13	AD_VP	NULL	100	90
114	11000.00	Den	Li	DLI	1.515.555.0114	2012-12-07	PU_MAN	NULL	100	30
120	8000.00	Matthew	Weiss	MWEISS	1.650.555.0120	2014-07-18	ST_MAN	NULL	100	50
121	8200.00	Adam	Fripp	AFRIPP	1.650.555.0121	2015-04-10	ST_MAN	NULL	100	50
122	7900.00	Payam	Kaufling	PKAUFLIN	1.650.555.0122	2013-05-01	ST_MAN	NULL	100	50
123	6500.00	Shanta	Vollman	SVOLLMAN	1.650.555.0123	2015-10-10	ST_MAN	NULL	100	50
124	5800.00	Kevin	Mourgos	KMOURGOS	1.650.555.0124	2017-11-16	ST_MAN	NULL	100	50
145	14000.00	John	Singh	JSINGH	44.1632.960000	2014-10-01	SA_MAN	0.40	100	80
146	13500.00	Karen	Partners	KPARTNER	44.1632.960001	2015-01-05	SA_MAN	0.30	100	80
147	12000.00	Alberto	Errazuriz	AERRAZUR	44.1632.960002	2015-03-10	SA_MAN	0.30	100	80
148	11000.00	Gerald	Cambraut	GCAMBRAU	44.1632.960003	2017-10-15	SA_MAN	0.30	100	80
149	10500.00	Eleni	Zlotkey	EZLOTKEY	44.1632.960004	2018-01-29	SA_MAN	0.20	100	80
201	13000.00	Michael	Martinez	MMARTINE	1.515.555.0166	2014-02-17	MK_MAN	NULL	100	20
125	3200.00	Julia	Nayer	JNAYER	1.650.555.0125	2015-07-16	ST_CLERK	NULL	120	50
126	2700.00	Irene	Mikkilineni	IMIKKILI	1.650.555.0126	2016-09-28	ST_CLERK	NULL	120	50
127	2400.00	James	Landry	JLANDRY	1.650.555.0127	2017-01-14	ST_CLERK	NULL	120	50
128	2200.00	Steven	Markle	SMARKLE	1.650.555.0128	2018-03-08	ST_CLERK	NULL	120	50
180	3200.00	Winston	Taylor	WTAYLOR	1.650.555.0145	2016-01-24	SH_CLERK	NULL	120	50
181	3100.00	Jean	Fleaur	JFLEAUR	1.650.555.0146	2016-02-23	SH_CLERK	NULL	120	50
182	2500.00	Martha	Sullivan	MSULLIVA	1.650.555.0147	2017-06-21	SH_CLERK	NULL	120	50
183	2800.00	Girard	Geoni	GGEONI	1.650.555.0148	2018-02-03	SH_CLERK	NULL	120	50

```

22 rows in set (0.02 sec)

```

11. Display the unique country_id from locations table.

SELECT DISTINCT COUNTRY_ID FROM LOCATIONS;

```

W2_99546_Pratiksha>SELECT DISTINCT COUNTRY_ID FROM LOCATIONS;

```

COUNTRY_ID
AU
BR
CA
CH
CN
DE
GB
IN
IT
JP
MX
NL
SG
US

```

14 rows in set (0.01 sec)

```

12. Display all employees whose have the job_id as IT_PROG and FI_ACCOUNT.

```

SELECT EMPLOYEE_ID, SALARY, FIRST_NAME, LAST_NAME, EMAIL, PHONE_NUMBER, HIRE_DATE,
JOB_ID, COMMISSION_PCT, MANAGER_ID, DEPARTMENT_ID

FROM employees

WHERE JOB_ID IN('IT_PROG' , 'FI_ACCOUNT' );

```

```
W2_99546_Pratiksha>SELECT EMPLOYEE_ID, SALARY, FIRST_NAME, LAST_NAME, EMAIL, PHONE_NUMBER, HIRE_DATE, JOB_ID, COMMISSION_PCT, MANAGER_ID, DEPARTMENT_ID
-> FROM employees
-> WHERE JOB_ID IN('IT_PROG' , 'FI_ACCOUNT' );
```

EMPLOYEE_ID	SALARY	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
109	9000.00	Daniel	Faviet	DFAVIET	1.515.555.0109	2012-08-16	FI_ACCOUNT	NULL	108	100
110	8200.00	John	Chen	JCHEN	1.515.555.0110	2015-09-28	FI_ACCOUNT	NULL	108	100
111	7700.00	Ismael	Sciarra	ISCIARRA	1.515.555.0111	2015-09-30	FI_ACCOUNT	NULL	108	100
112	7800.00	Jose Manuel	Urman	JMURMAN	1.515.555.0112	2016-03-07	FI_ACCOUNT	NULL	108	100
113	6900.00	Luis	Popp	LPOPP	1.515.555.0113	2017-12-07	FI_ACCOUNT	NULL	108	100
103	9000.00	Alexander	James	AJAMES	1.590.555.0103	2016-01-03	IT_PROG	NULL	102	60
104	6000.00	Bruce	Miller	BMILLER	1.590.555.0104	2017-05-21	IT_PROG	NULL	103	60
105	4800.00	David	Williams	DWILLIAMS	1.590.555.0105	2015-06-25	IT_PROG	NULL	103	60
106	4800.00	Valli	Jackson	VJACKSON	1.590.555.0106	2016-02-05	IT_PROG	NULL	103	60
107	4200.00	Diana	Nguyen	DNGUYEN	1.590.555.0107	2017-02-07	IT_PROG	NULL	103	60

10 rows in set (0.00 sec)

13. Display all countries in ascending order.

```
SELECT COUNTRY_ID, COUNTRY_NAME, REGION_ID
FROM COUNTRIES
ORDER BY COUNTRY_NAME ;
```

```
W2_99546_Pratiksha>SELECT COUNTRY_ID, COUNTRY_NAME, REGION_ID
-> FROM COUNTRIES
-> ORDER BY COUNTRY_NAME ;
```

COUNTRY_ID	COUNTRY_NAME	REGION_ID
AR	Argentina	20
AU	Australia	40
BE	Belgium	10
BR	Brazil	20
CA	Canada	20
CN	China	30
DK	Denmark	10
EG	Egypt	50
FR	France	10
DE	Germany	10
IN	India	30
IL	Israel	30
IT	Italy	10
JP	Japan	30
KW	Kuwait	30
ML	Malaysia	30
MX	Mexico	20
NL	Netherlands	10
NG	Nigeria	50
SG	Singapore	30
CH	Switzerland	10
GB	United Kingdom of Great Britain and Nort	10
US	United States of America	20
ZM	Zambia	50
ZW	Zimbabwe	50

25 rows in set (0.01 sec)